

Landscape-scale population connectivity in two parasitoid species associated with the spruce budworm: Testing the birdfeeder effect using genetic data

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Abstract

Periodic and spatially synchronous outbreaks of insect pests have dramatic consequences for boreal and sub-boreal forests. Within these multitrophic systems, parasitoids can be stabilizing agents by dispersing toward patches containing higher host density (the so-called birdfeeder effect). However, we know little about the dispersal abilities of parasitoids in continuous forested landscapes, limiting our understanding of the spatiotemporal dynamics of host–parasitoid systems, and constraining our ability to predict forest resilience in the context of global changes. In this study, we investigate the spatial genetic structure and spatial variation in genetic diversity of two important species of spruce budworm larval parasitoids during outbreaks: *Apanteles fumiferanae* Viereck (Braconidae) and *Glypta fumiferanae* (Viereck) (Ichneumonidae). Using parasitoids sampled in 2014 from 26 and 29 locations across a study area of 350,000 km², we identified 1,012 and 992 neutral SNP loci for *A. fumiferanae* ($N = 279$ individuals) and *G. fumiferanae* ($N = 382$), respectively. Using DAPC, PCA, AMOVA, and IBD analyses, we found evidence for panmixia and high genetic connectivity for both species, matching the previously described genetic structure of the spruce budworm within the same context, suggesting similar effective dispersal during outbreaks and high parasitoid population densities between outbreaks. We also found a significant negative relationship between genetic diversity and latitude for *A. fumiferanae* but not for *G. fumiferanae*, suggesting that northern range limits may vary by species within the spruce budworm parasitoid community. These spatial dynamics should be considered when predicting future insect outbreak severities in boreal landscapes.

KEYWORDS

connectivity, dispersal, genetic structure, insect outbreaks, parasitoids, spruce budworm

1 | INTRODUCTION

Identification and characterization of the mechanisms that maintain ecosystem stability and resilience are needed to ensure the sustainability of ecosystem services in a changing world (Hisano et al.,

2018; Oliver et al., 2015). One such mechanism influencing ecosystem services is predation (McCann et al., 2005). Intuitively, higher-order predators are larger than their prey and tend to be more mobile (Ripple et al., 2014). Predators at higher trophic levels have the capacity to modulate their foraging behaviour (numerical and

functional responses) as a function of local prey densities, which can, in turn, attenuate food web fluctuations (McCann & Rooney, 2009; Rooney et al., 2008). As a result of such effects on food web dynamics, apex predators have been shown to indirectly influence primary productivity (Ripple et al., 2014). Well-known examples include wolves (Beschta & Ripple, 2009), dingoes (Morris & Letnic, 2017), cougars (Ripple & Beschta, 2008), and grizzly bears (Armstrong et al., 2016).

Despite the known role of large, mobile predators in stabilizing ecosystems through trophic cascades (McCann & Rooney, 2009; Rooney et al., 2008), these processes do not necessarily apply to insect-parasitoid systems. Insect parasitoids can be considered apex predators in almost all insect communities (Godfray, 1994; Hawkins, 1994). Their larvae develop on, or within, a single insect host (Godfray, 1994). As a result, adult parasitoids have smaller body sizes than their hosts, with sometimes lower dispersal abilities than their host (e.g., van Nouhuys and Hanski (2002)). However, other mechanisms such as longevity, response to host-related chemical cues, and host attachment for dispersal can facilitate parasitoid movement over large areas (McCann & Rooney, 2009). Improving our understanding of parasitoid movement and population connectivity is essential to characterizing their role in the spatial outbreak dynamics of forest insect pests and in maintaining ecosystem stability and resilience.

In boreal and subboreal regions, outbreaks of native forest insect pests such as the mountain pine beetle (*Dendroctonus ponderosae* (Hopkins) (Coleoptera: Curculionidae)), the forest tent caterpillar (*Malacosoma disstria* Hübner (Lepidoptera: Lasiocampidae)), the hemlock looper (*Lambdina fiscellaria* Guenée (Lepidoptera: Geometridae)), and the spruce budworm (*Choristoneura fumiferanae* (Clemens) (Lepidoptera: Tortricidae)), have dramatic ecological and economic consequences (Chang et al., 2012; Chen, 2014; Hébert et al., 2001; Peltonen et al., 2002; Pureswaran et al., 2018; Schowalter, 2017; Williams & Birdsey, 2003). In Eastern North America, the spruce budworm is the most important defoliator of balsam fir (*Abies balsamea*) and spruce (*Picea glauca*, *P. rubens*, and *P. mariana*) trees (Sturtevant et al., 2015; Volney & Fleming, 2007). Large-scale, spatially synchronous population oscillations (i.e., outbreaks; Figure 1) have occurred at 30–40 year intervals over the past millennia (Boulanger & Arseneault, 2004; Jardon et al., 2003; Simard et al., 2006). Outbreaks typically persist for 10–15 years and result in widespread damage and mortality to fir and spruce trees (Berguet et al., 2021; Ostaf & MacLean, 1989). Outbreaks also negatively impact timber flow (Chang et al., 2012), can increase the risk of wildfire (James et al., 2017), and reduce rates of forest carbon uptake (Dymond et al., 2010).

Because of these diverse adverse effects, the spatiotemporal dynamics of spruce budworm populations have been the subject of intense ecological inquiry and debate over the last century (Pureswaran et al., 2016). The budworm's cyclic and typically spatially synchronous dynamics involve a complex interplay among (i) spatial correlation in environmental attributes such as temperature and precipitations (i.e., the Moran effect; Koenig, 2002; Moran,

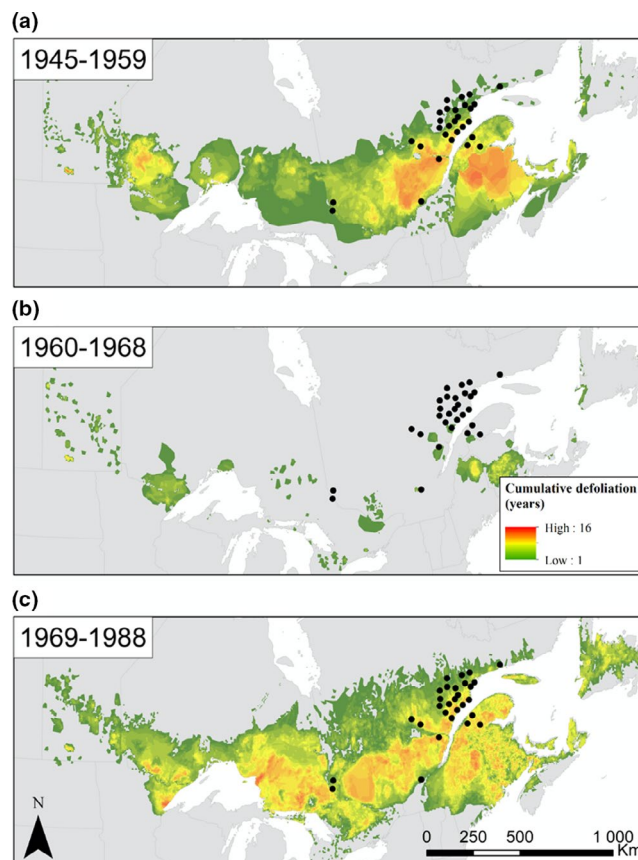


FIGURE 1 Extent of past spruce budworm outbreaks. Spruce budworm cumulative defoliation from (a) 1945 to 1959, (b) 1960 to 1968, and (c) 1969 to 1988. Cumulative defoliation maps have been compiled using the digital data set of Williams and Birdsey (2003), and pixel colours indicate the cumulative number of years (from 1 to 16) for which defoliation has been detected. Maps use a binomial classification (presence-absence), regardless of defoliation severity. Dots (•) show the locations of sites sampled in 2014 [Colour figure can be viewed at wileyonlinelibrary.com]

1953), (ii) dispersal from areas with high population densities (epi-centres) through mass exodus flights that colonize new, distant areas with low population densities (Boulanger et al., 2017; James et al., 2015; Larroque et al., 2019; Williams & Liebhold, 2000), and (iii) natural enemies (predators and parasitoids) that modulate their foraging behaviour as a function of the local densities of their hosts, which contributes to the regional synchronization of host densities (Eveleigh et al., 2007; McCann & Rooney, 2009). Recent work has shown that dispersal is a key mechanism involved in driving population synchrony of spruce budworm outbreaks (Larroque et al., 2019). However, no consensus has yet been reached regarding the relative importance of each of the three main ecological mechanisms in explaining and predicting irruptive population dynamics in the spruce budworm system (Pureswaran et al., 2016; Royama, 1984; Sturtevant et al., 2015). Our objective in this paper is to explore in greater detail the role of parasitoids in driving large-scale, spatially synchronous population outbreaks.

Mortality rates of spruce budworm larvae and pupae are influenced by a diverse community of hymenopteran and dipteran

parasitoids (Eveleigh et al., 2007; Marrec et al., 2018; Royama et al., 2017; Smith et al., 2011). This community has long been thought to behave like a delayed, density-dependent single predator (Royama, 1984). Its food web structure has been shown to change locally as a function of local spruce budworm density (Miller & Renault, 1976) or by means of the birdfeeder effect, whereby generalist consumers exhibit prey switching behaviour and move throughout landscapes toward areas exhibiting higher host resources (Eveleigh et al., 2007). The existence and the potential efficacy of the birdfeeder effect over large spatial scales thus mainly depend on the host range and the dispersal capability of parasitoids relative to their hosts. Indeed, if generalist parasitoids have dispersal capabilities similar to (or greater than) their hosts, the birdfeeder effect would contribute to broad-scale outbreak synchronization, because the parasitoids would be able to move rapidly from low to high host-density areas (Cronin & Reeve, 2005; Eveleigh et al., 2007; Ims & Steen, 1990; McCann & Rooney, 2009; Peltonen et al., 2002; Ydenberg, 1987). Unfortunately, the dispersal abilities of spruce budworm parasitoids remain poorly characterized. Traditional field methods based on recapture rate measurements are difficult to apply to small organisms such as insect parasitoids. Consequently, we know very little about (i) at which spatial scale(s) the birdfeeder effect occurs and its potential role in synchronizing spruce budworm outbreaks over large geographic areas, and (ii) whether the strength of the birdfeeder effect varies geographically (e.g., in response to climatic and vegetational diversity gradients).

Estimates of spatial genetic structure can be used to indirectly infer the rate and scale of effective dispersal, quantify population connectivity, and identify barriers to movement (Broquet & Petit, 2009). If parasitoid populations have greater dispersal capacities than that of their hosts, one would expect to observe relatively lower genetic differentiation. For example, the highly dispersive parasitoid wasp *Hyposoter horticola* Gravenhost (Ichneumonidae) exhibits less spatial genetic structure than its butterfly host, the Glanville fritillary butterfly, *Melitaea cinxia* L. (Nymphalidae) (Couchoux et al., 2016). Alternatively, if parasitoid dispersal is weaker than that of its host, then relatively higher genetic differentiation is expected. Examples include *Cotesia melitaearum* (Wilkinson) (Hymenoptera: Braconidae), which also parasitizes the Glanville fritillary butterfly (Kankare et al., 2005), *Eurytoma robusta* Mayr (Chalcididae) attacking *Urophora cardui* (L.) (Diptera: Tephritidae) (Johannsen & Seitz, 2003), and *Neotypus melanocephalus* Gmelin (Ichneumonidae) attacking the endangered butterfly *Maculinea nausithous* Bergtässer (Lycaenidae) (Anton et al., 2007). Comparing the spatial genetic structure of the spruce budworm, described recently in Larroque et al. (2019), with the spatial genetic structure of its generalists versus specialists parasitoids could therefore shed light on the relative dispersal abilities of these species. This information could provide the basis for an assessment of whether spruce budworm parasitoids contribute to stabilizing spatial outbreak dynamics through the birdfeeder effect.

A description of spatial variation in parasitoid genetic diversity may also contribute to a better understanding of future spruce

budworm outbreak dynamics in the context of climate change. In response to global climate warming, the climatic envelope suitable to the spruce budworm is expected to shift northward, where defoliation and mortality of coniferous stands were minor during past outbreaks (Figure 1) (Gray, 2008; Pureswaran et al., 2015; Régnière et al., 2012). Northern forests have reduced tree-species diversity relative to forests in the central and southern part of the spruce budworm's range (Legault & James, 2018; Qian & Ricklefs, 2007). Although boreal forest ecosystems can show high stability (Stralberg et al., 2020), vegetational diversity is generally thought to be associated with ecological mechanisms that promote ecosystem stability and resilience to disturbances (Jactel et al., 2017, 2021; Loreau & de Mazancourt, 2013; McCann, 2000). Tree-species diversity has been shown to reduce damage caused by specialist herbivores – such as the spruce budworm – in mixed stands with phylogenetically distant tree species, in part through top-down effects of generalist natural enemies that benefit from increased food resources and shelters (Jactel et al., 2021). Parasitism from spruce budworm generalist parasitoids has indeed been shown to be higher in diverse forests during outbreaks (Cappuccino et al., 1998; Eveleigh et al., 2007; Legault & James, 2018; Quayle et al., 2003). As a result, overall parasitoid densities may be lower at higher latitudes, with potential consequences for future outbreak dynamics.

Because spruce budworm outbreaks increase parasitoid population densities in affected regions (Miller, 1959, 1960), we hypothesize that many species of spruce budworm parasitoid follow an abundant-center distribution (*sensu* Sagarin & Gaines, 2002) that overlaps with the spruce budworm outbreak range (Figure 1). Specifically, we expect parasitoid subpopulations in northern parts of the spruce budworm range to be less well connected than central populations within the central part of the range. This reduced connectivity may lead to an inefficient birdfeeder effect in northern landscapes and unpredictable spruce budworm outbreak dynamics in the future. The population genetic consequence of an abundant-centre population distribution can be described using the central-marginal hypothesis (Eckert et al., 2008). The central-marginal hypothesis is the genetic conceptual analogue to the demographic abundant-centre model. Specifically, the central-marginal hypothesis suggests that northern parasitoid populations should display greater genetic differentiation and lower genetic diversity than populations from the center of the spruce budworm range (Eckert et al., 2008).

In this study, we describe for the first time the spatial genetic structure of two common hymenopteran spruce budworm parasitoids: *Apanteles fumiferanae* Viereck (Braconidae) and *Glypta fumiferanae* (Viereck) (Ichneumonidae). We assessed their metapopulation structure and described the genetic connectivity between populations from different spruce budworm outbreak areas. Outbreak areas are defined as geographically distinct zones where spruce budworm defoliation has been detected for several years by the Quebec' Ministère des Forêts, de la Faune et des Parcs (Legault & James, 2018; Ministère des Forêts de la Faune et des Parcs, 2014; Figure 2a). These outbreak areas are located within the core and at the northern periphery of the spruce budworm historic outbreak

range (Figure 1), which also follows a latitudinal gradient of decreasing forest diversity (Figure 2b) and increasing spruce budworm host tree density (Figure 2c) (Bouchard & Auger, 2014; Legault & James, 2018; Pureswaran et al., 2015). We hypothesize that (i) recent spruce budworm outbreak dynamics shaped the contemporary genetic structure of *A. fumiferanae* and *G. fumiferanae*. That is, we expect to find clusters of parasitoid genotypes that match the different outbreak areas we sampled; Figure 2a) and (ii) parasitoid genetic connectivity and diversity decreases with increasing latitude as we approach the northern limits of the spruce budworm outbreaks range (Figure 1). Identifying how genetic diversity and connectivity of parasitoids vary with latitude is important both for the theoretical

ecology of cycling populations in continuous forest landscapes and for improving our understanding of the drivers of large-scale insect outbreaks.

2 | MATERIALS AND METHODS

2.1 | Study organisms

The spruce budworm parasitoid community is diverse: up to ~100 species attack spruce budworm eggs, larvae, or pupae (Eveleigh et al., 2007). However, most parasitism is attributed to a few key species (Blais, 1960; Eveleigh & Johns, 2014; Nealis, 2016). This study focuses on *A. fumiferanae* and *G. fumiferanae*, the two most common larval parasitoids of *Choristoneura* species across North America during outbreaks (Blais, 1960; Henry, 1996; Legault & James, 2018; Marrec et al., 2018; McGugan & Blais, 1959; Nealis, 1991; Niwa et al., 1987; Royama et al., 2017; Schaupp et al., 1991). Both species are solitary and univoltine (Brown, 1946a, 1946b). They attack *Choristoneura* larvae in late summer, and their progeny overwinter as eggs or small larvae (Miller, 1959, 1960) within second instar hosts (Lewis, 1960; McLeod, 1977). Immature parasitoids resume development as their hosts become active the following spring (Lewis, 1960) until they egress from late larval instars (McGugan, 1955). Both species have been shown to be well synchronized with their hosts, despite yearly variations in weather (Nealis, 1988; Nyrop & Simmons, 1986). The longevity of adult females is similar for both species (Elliott et al., 1987), suggesting that this trait probably does not affect their relative annual dispersal distances.

Parasitism by *A. fumiferanae* and *G. fumiferanae* has been well documented during past and present spruce budworm outbreaks in Eastern Canada (Blais, 1960, 1965; Eveleigh & Johns, 2014; Legault & James, 2018; McGugan, 1955; McLeod, 1977; Miller & Renault, 1976; Royama et al., 2017). Parasitism rates are variable, but at the peak of the last major outbreak, during the 1970s, they were consistently high across the central part of the province of Québec in 1975 and 1976, reaching 56% and 24%, for *A. fumiferanae* and *G. fumiferanae*, respectively (McLeod, 1977). However, the outbreak analysed in this study (since ~2006) is occurring mostly at higher latitudes (Pureswaran et al., 2015), where parasitism by *A. fumiferanae* and *G. fumiferanae* is lower than it was in previous outbreaks (Legault & James, 2018). These reduced parasitism rates suggest that northern parasitoid populations might be approaching their range limits (Eckert et al., 2008).

Even though *A. fumiferanae* and *G. fumiferanae* have very similar life cycles, the frequency of multiparasitism involving both species during epidemic periods is low (McLeod, 1977), suggesting that *G. fumiferanae* females select hosts unparasitized by *A. fumiferanae*. Indeed, *G. fumiferanae* has been suggested to be an inferior interspecific competitor relative to *A. fumiferanae* (Lewis, 1960), which could push *G. fumiferanae* to disperse further in landscapes under a potential competition-colonization trade-off scenario (Amarasekare, 2003). During endemic periods, parasitism rates by *A. fumiferanae*

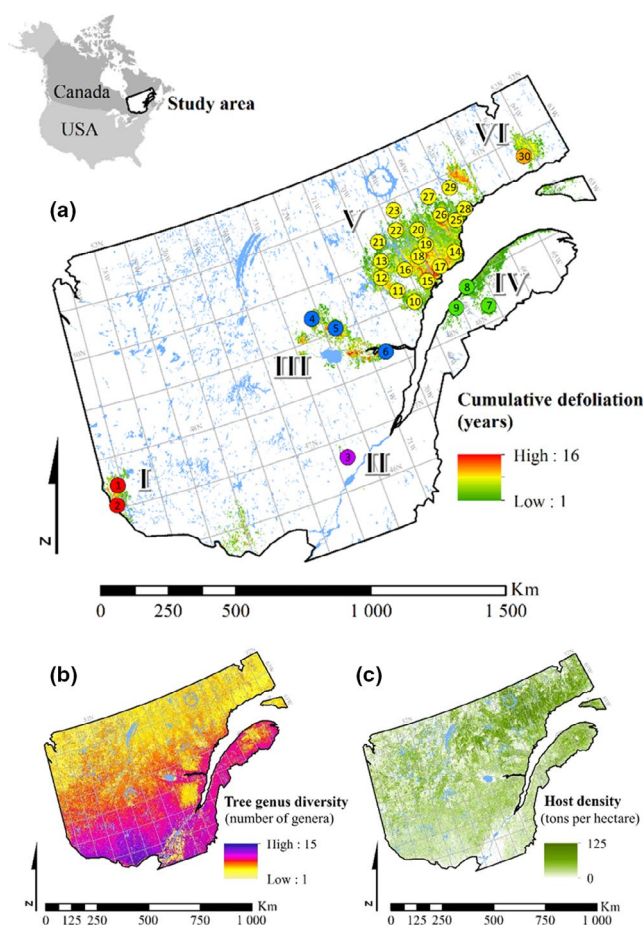


FIGURE 2 Study area. (a) Study area in continental context showing the locations of the 29 sampling sites within the six outbreak areas (Roman numerals) in the province of Québec (Canada) with extents of spruce budworm cumulative defoliation (i.e., the cumulative number of years [from 1 to 16] for which defoliation has been detected) as of 2014. Different site colours represent different outbreak areas (see Table 1). (b, c) Maps of spatial variation in tree genus diversity (number of tree genera detected in each pixel) and spruce budworm host tree (balsam fir and spruce) density (tons per hectare) over the extent of the study area. Data are from Beaudoin et al. (2014) and the Ministère des Forêts, de la Faune et des Parcs du Québec (see text and Legault & James [2018] for details) [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

TABLE 1 Geographic coordinates of each site sampled in 2014, as well as the number of female individuals genotyped for each species (N), expected heterozygosity (H_e), and mean allelic richness (A_r). Outbreak areas and site numbers and colours are as in Figure 2a. Populations represented by a sample size of one (1) individual were not used in genetic analyses using allele frequencies at the site level [Colour table can be viewed at wileyonlinelibrary.com]

Outbreak area	Site	Longitude	Latitude	<i>A. fumiferanae</i>			<i>G. fumiferanae</i>		
				N	H_e	A_r	N	H_e	A_r
I. Abitibi-Témiscamingue	1 ●	-79.11	47.42	22	0.3028	1.6699	7	0.2852	1.6462
	2 ●	-79.27	47.05	20	0.3081	1.6828	18	0.2917	1.6547
II. Mauricie	3 ●	-72.65	46.62	15	0.3034	1.6629	15	0.2936	1.6583
III. Saguenay-Lac-St-Jean	4 ●	-72.23	49.40	10	0.2911	1.6533	14	0.2906	1.6531
	5 ●	-71.66	49.06	18	0.3016	1.6643	15	0.2909	1.6532
	6 ●	-70.54	48.28	22	0.3074	1.6639	14	0.2925	1.6563
IV. Bas-Saint-Laurent	7 ●	-68.14	48.52	20	0.3040	1.6632	10	0.2818	1.6420
	8 ●	-67.60	48.81	23	0.3046	1.6588	7	0.2789	1.6348
	9 ●	-67.24	48.30	18	0.3061	1.6725	7	0.2852	1.6425
V. Côte-Nord	10 ●	-69.19	48.96	12	0.3025	1.6719	15	0.2957	1.6635
	11 ●	-69.53	49.28	11	0.2962	1.6628	14	0.2928	1.6557
	12 ●	-69.84	49.65	11	0.2928	1.6525	14	0.2964	1.6657
	13 ●	-69.64	49.95	12	0.2945	1.6460	15	0.2899	1.6488
	14 ●	-67.49	49.54	23	0.3026	1.6678	15	0.2932	1.6587
	15 ●	-68.60	49.24	7	0.2840	1.6416	15	0.2938	1.6604
	16 ●	-69.08	49.61	3	0.2635	1.6096	14	0.2913	1.6540
	17 ●	-68.07	49.39	3	0.2571	1.5930	15	0.2974	1.6665
	18 ●	-68.54	49.75	(1)	-	-	15	0.2899	1.6512
	19 ●	-68.20	49.90	9	0.2929	1.6588	11	0.2864	1.6457
	20 ●	-68.24	50.23	7	0.2938	1.6623	15	0.2914	1.6550
	21 ●	-69.49	50.32	(1)	-	-	15	0.2923	1.6547
	22 ●	-68.85	50.39	(1)	-	-	12	0.2847	1.6410
	23 ●	-68.66	50.77	5	0.2727	1.6165	15	0.2920	1.6563
	25 ●	-67.05	50.09	(1)	-	-	15	0.2959	1.6627
	26 ●	-67.40	50.31	-	-	-	14	0.2935	1.6593
	27 ●	-67.45	50.72	(1)	-	-	(1)	-	-
	28 ●	-66.66	50.22	-	-	-	15	0.2978	1.6684
	29 ●	-66.80	50.72	3	0.2690	1.5634	15	0.2941	1.6614
VI. Basse-Côte-Nord	30 ●	-64.30	50.62	-	-	-	15	0.2938	1.6582

are comparable to those obtained from outbreak records, whereas parasitism by *G. fumiferanae* is almost null (Miller & Renault, 1976). Thus, *A. fumiferanae* and *G. fumiferanae* may have different host ranges when spruce budworm populations are very low (Miller & Renault, 1976), which may limit interspecific competition (Lewis, 1960) when this food source is scarce and further influence their contemporary genetic structure. In addition to these historical records, DNA barcoding of parasitoids reared from different lepidopteran hosts collected on balsam fir foliage during a spruce budworm outbreak suggest that *G. fumiferanae* can be considered a host specialist, whereas *A. fumiferanae* can be considered a host generalist. However, Smith et al. (2011) barcoded *G. fumiferanae* from only two other hosts than the spruce budworm, and both from balsam fir trees, raising the possibility that both *A. fumiferanae* and *G. fumiferanae* may be host generalists. Taken together, these results suggest

that *A. fumiferanae* may be an effective host switcher that is able to maintain relatively large population sizes when spruce budworm populations are low (Miller & Renault, 1976). *Glypta fumiferanae*, by contrast, does not appear to use this ecological strategy, and could persist across the landscape through higher dispersal abilities (Amarasekare, 2003; Lewis, 1960).

2.2 | Parasitoid sampling

Spruce budworm larvae were collected from late May to mid-July 2014 from 29 sites located within and around six distinct outbreak areas (i.e., areas experiencing significant spruce budworm defoliation) in the province of Québec, Canada (Figure 2a; Table 1). Because the number of man-hours needed to sample *A. fumiferanae* and *G.*

fumiferanae individuals when spruce budworm densities are very low is extremely high (Blais, 1965; Miller & Renault, 1976), we did not sample in areas where defoliation was absent. Côte-Nord, by far the largest region sampled in this study, is heavily forested and more affected, in terms of defoliated area, by the current spruce budworm outbreak, which started around 2006 (outbreak areas V and IV); this explains the more extensive sampling effort. At each site, we sampled ~1,000 spruce budworm larvae by randomly collecting foliage of the lower and mid-crown of at least 10 defoliated balsam fir and spruce (*P. glauca* and/or *P. mariana*) trees using a pole pruner. All sampled trees were 5–10 m apart to decrease the probability of collecting closely related parasitoids. Foliage was brought back to the laboratory in large paper bags where larvae were extracted by hand and reared individually on artificial diet (Hervet et al., 2016) in 1-oz. plastic cups at 20°C until budworm pupation or egression of parasitoid larvae. Parasitoid cocoons were then transferred to clean 1-oz. plastic cups until emergence of adults. All adult parasitoid specimens were starved for two to three days (to reduce potential amounts of host DNA in their gut), sexed, and identified to species using morphological taxonomic keys (Bennett, 2008; Fernández-Triana & Huber, 2010). Specimens were then stored individually in dry conditions at –80°C prior to DNA extraction.

2.3 | DNA extraction

Because both *A. fumiferanae* and *G. fumiferanae* are haplodiploid (Shon & Shea, 1976), only female wasps were used for genetic analyses based on measures of heterozygosity. Total genomic DNA was extracted from all available reared individuals (1–23 females per species per site; Table 1) using the DNeasy Blood and Tissue Kit (Qiagen). For *A. fumiferanae*, we used a modified version of the extraction protocol to increase final DNA concentrations, in which only 20 µl of AE buffer was pipetted on the silica-based membrane during elution, with an incubation time of 30 min before each elution. The resulting eluate was then reused as AE buffer for three more elutions by pipetting the buffer each time on different areas of the membrane. DNA concentrations and purity ratios were obtained using spectrophotometry (NanoDrop 1000, Techno Scientific). Only samples with 260/280 nm absorbance ratios >1.6 and 260/230 ratios >0.95 were retained for genotyping-by-sequencing. Subsequently, 200 ng of DNA from each sample of *G. fumiferanae* and 150 ng from each sample of *A. fumiferanae* were stored at –80°C in a 96-well format prior to sequencing.

2.4 | Genotyping-by-sequencing

Genotyping-by-sequencing libraries were constructed using the protocol of (Poland et al., 2012) at the Institut de Biologie Intégrative et des Systèmes (IBIS; Université Laval, Québec City, QC, Canada). DNA was digested using high-fidelity PstI and MspI restriction enzymes (New England Biolabs). This combination of restriction enzymes was chosen to maximize data compatibility with other

projects on spruce budworm population genetics (Larroque et al., 2019, 2020). Each sample in a 96-well plate received a unique barcode adapter (Table S1 of Elshire et al., 2011) that was ligated on the PstI cut site and a common adapter ligated on the MspI cut site. For each 96-well plate, samples were multiplexed and adapter-DNA combinations were PCR-amplified (18 cycles; Elshire et al., 2011). Libraries (96 samples each) were then single-end sequenced (100 bp reads) using an Illumina HiSeq2000 (Illumina Inc.) at the McGill University and Génome Québec Innovation Centre (Montreal, QC, Canada). Base calling accuracy measured by the Phred quality score (Q score) was between 34 and 35 for all libraries.

2.5 | SNP calling and filtering

Because there is no reference genome available for *A. fumiferanae* and *G. fumiferanae*, we used the TASSEL 3.0 UNEAK (Universal Network-Enabled Analysis Kit) pipeline for single-nucleotide-polymorphism (SNP) calling (Lu et al., 2013). Instead of aligning sequences to a reference genome, the UNEAK pipeline proceeds to pairwise alignments of sequences from all samples, leading to the formation of networks, where a one-bp mismatch is considered a candidate SNP (Lu et al., 2013). During these procedures, only sequences with five or more reads over all individuals were kept, creating a master list of sequences which serves as a “pseudo reference genome”. During alignment, a 0.03 error tolerance rate (default) was used (Lu et al., 2013). SNP loci with a minor allele frequency (MAF) <5% were removed (Marees et al., 2018). Genotypes in the resulting HapMap file were corrected using the program Uneak_filter developed by IBIS (Table S1) to consider low allele frequencies that could have resulted from sequencing errors. Loci absent in >25% of individuals and individuals with >10% missing loci were pruned out of the data matrix using TASSEL 5 (Bradbury et al., 2007). We then created a pruned data set of SNP loci for each species by removing SNPs in high linkage disequilibrium (LD) using the snpgdsLDpruning function of the R package SNPrelate (Zheng et al., 2012). SNPs in high LD with a correlation coefficient ≥0.3 were dropped (Table S2) to remove highly correlated variants that add negligible extra information and can excessively impact downstream population genetic analyses, such as principal component analyses (PCA) (Zou et al., 2010).

Finally, because this study focuses mainly on parasitoid dispersal, we filtered out SNPs that were identified as putatively under selection by a combination of the following methods: PCAdapt (Luu et al., 2017), sNMF (Frichot et al., 2014; Martins et al., 2016), and OutFlank (Whitlock & Lotterhos, 2015). Loci identified as putatively under selection by at least one method (Table S3) were removed from data sets prior to undertaking subsequent analyses.

2.6 | Clustering and spatial genetic variation

Using the filtered SNP data sets, we first determined how many genetic clusters were useful for describing each parasitoid species and

whether spatial connectivity between identified clusters was associated with different outbreak areas (Figure 2a). We used discriminant analysis of principal component (DAPC) functions implemented in the *adeigenet* (Jombart & Ahmed, 2011) R package to identify genetic groups of individuals using *k-means* clustering algorithms. This approach works without a priori assumptions on population subdivisions or population genetic models (Jombart et al., 2010). To assess the influence of outbreak areas on genetic structure, we partitioned genetic variability for each species using analysis of molecular variance (AMOVA) implemented in the R package *pegas* (Paradis, 2010). Hierarchical AMOVAs were performed using 999 permutations and outbreak areas and sampling sites within outbreak areas as grouping factors (Figure 2a, Table 1). Using the R package *adeigenet* (Jombart & Ahmed, 2011), we also performed PCAs using outbreak areas as a grouping factor for *A. fumiferanae* and *G. fumiferanae* individuals.

2.7 | Isolation by distance

If parasitoid populations are not well connected through dispersal within our study area, genetic and geographic distances between pairs of parasitoid individuals are expected to be correlated (i.e., isolation by distance [IBD]; Wright, 1943). For each parasitoid species, interindividual genetic distances were calculated using PCA: we created multiple principal components (PCs) with a matrix of allele occurrence (0, 1, or 2) using *adeigenet* (Jombart & Ahmed, 2011) and then derived a distance matrix from the Euclidean distances between individuals in the multidimensional space created by the first 64 PC axes (Larroque et al., 2019; Shirk et al., 2017). We then tested for IBD by testing the correlation between genetic distances and geographic distances (in km) between all pairs of individuals. Statistical significance of these relationships was evaluated with Mantel tests (9,999 permutations) using the *ade4* package (Dray & Dufour, 2007) in R. Although the Mantel test is an imperfect tool to test for IBD (Legendre et al., 2015), we choose this approach for comparison purposes with recent IBD analyses on the spruce budworm itself (Larroque et al., 2019). However, distance-based Moran's eigenvector mapping (db-MEM) were also performed on the data sets using R package *MEMGENE* (function *mgQuick*) with 999 permutations for each pair of matrices (Galpern et al., 2014). This method allows to compare matrices of genetic and geographic distances without violating assumptions of linearity and homoscedasticity (Legendre et al., 2015).

2.8 | Genetic diversity

We tested the central-marginal hypothesis (Eckert et al., 2008) that genetic diversity and connectivity decrease with increasing latitude. For each species at each sample location, we calculated expected heterozygosity (H_e) using the function *Hs* as implemented in the R package *adeigenet* (Jombart & Ahmed, 2011) and allelic richness (A_r) using the function *allelic.rich* as implemented in the R package *PopGenReport* (Adamack & Gruber, 2014). Pearson correlations

between heterozygosity and latitude and between allelic richness and latitude were then performed for each parasitoid species. Because sample sizes varied between sampling sites, correlations were weighted by the number of genotyped individuals (N ; Table 1). Sampling sites with only one individual were eliminated before performing the correlations. To reduce potential bias related to low sample sizes in some sites, we also performed the correlations using only sites with at least 10 individuals genotyped.

3 | RESULTS

3.1 | Genotyping

During preliminary analyses (results not shown), we ran the UNEAK pipeline using all DNA sequences from both *A. fumiferanae* and *G. fumiferanae* individuals. We found no SNPs that were common to both species. If we had unintentionally sequenced spruce budworm DNA, we would expect to find common loci (i.e., spruce budworm SNPs) between our two parasitoid species. We are thus confident that we did not sequence spruce budworm DNA. Using the UNEAK pipeline on each parasitoid species separately, 19,890 SNPs were identified for *A. fumiferanae* and 19,555 for *G. fumiferanae* (Table S1). After filtering, 1,012 and 992 loci that met our stringent selection criteria (i.e., presence in at least 75% of the individuals, approximately in linkage equilibrium and putatively neutral) were retained for *A. fumiferanae* ($N = 279$ individuals) and *G. fumiferanae* ($N = 382$), respectively (see Table S3 for lists of loci classified as potentially under selection).

3.2 | Genetic clustering and spatial genetic variation

Using DAPC on the filtered SNP data sets, we did not find evidence of genetic clustering for either *A. fumiferanae* or *G. fumiferanae*. For each species, Bayesian information criterion (BIC) values were always minimized when assuming a single population (Jombart et al., 2010; Figure 3).

Using AMOVA, we found that more than 99.7% of the genetic variation for each parasitoid species was attributable to variation among individuals (Table 2). Although we found a significant amount of genetic variation among sites within outbreak areas for *G. fumiferanae*, the percentage of genetic variation was very low (0.254%). The lack of genetic structure associated with outbreak areas was particularly clear in our examination of PCA biplots (Figure 4), which showed no clear pattern of aggregation of individuals. Indeed, none of the axes were significant under a broken-stick model (results not shown). For *A. fumiferanae*, only 0.89% and 0.88% of the genetic variance between individuals was explained by the first and second axis of the PCAs, respectively (Figure 4a). For *G. fumiferanae*, the first and second axis explained only 0.77% and 0.71% of the variation among individuals, respectively (Figure 4b).

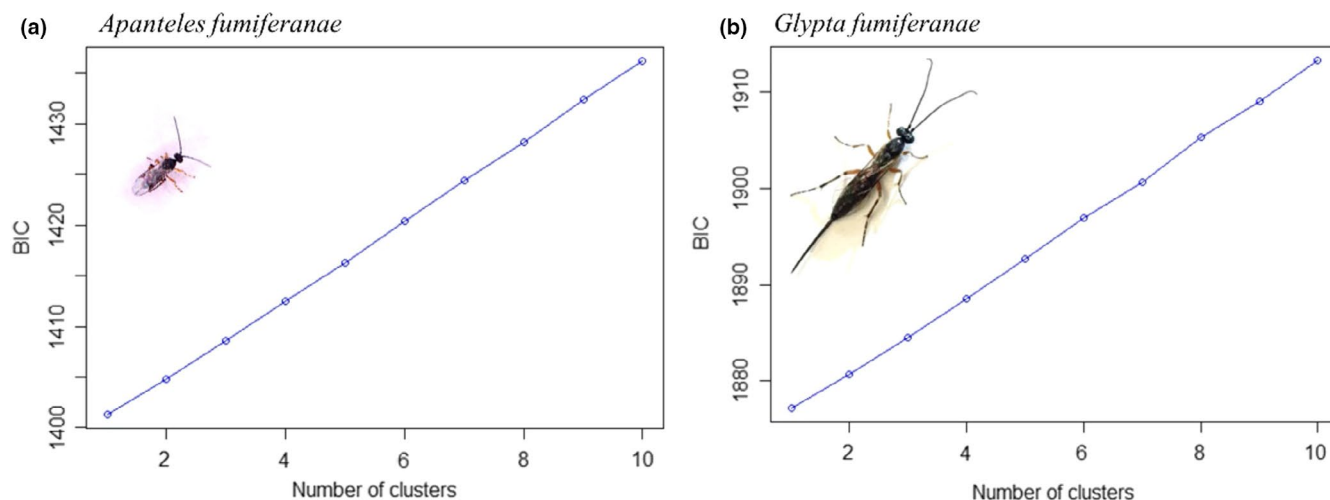
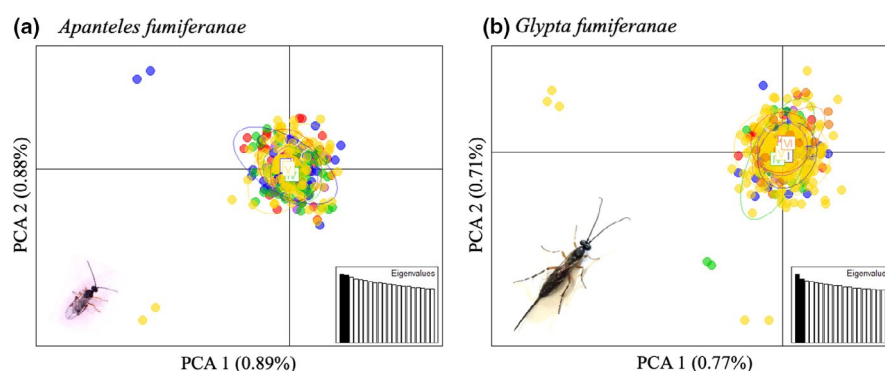


FIGURE 3 Inference of the number of clusters in genetic data sets. (a) *Apanteles fumiferanae*, (b) *Glypta fumiferanae*. For each species, the function *find.clusters* (Jombart et al., 2010) was used to identify the most useful number of clusters to describe genetic data sets. The Bayesian information criterion (BIC) is provided for up to 10 genetic clusters. The chosen number of clusters is one for each species because at this level of clustering BIC values are at the minimum (Jombart et al., 2010) [Colour figure can be viewed at wileyonlinelibrary.com]

TABLE 2 Results of AMOVAs for *Apanteles fumiferanae* and *Glypta fumiferanae* using outbreak areas and sampling sites (see Table 1 and Figure 2a) as grouping factors for the individuals

Species	Variance component	Variance (σ^2)	Variance (%)	p-value
<i>A. fumiferanae</i>	Among outbreak areas	0	0	0.845
	Among sites within outbreak areas	1.127	0.178	0.105
	Among individuals within sites	632.335	99.822	
<i>G. fumiferanae</i>	Among outbreak areas	0	0	0.955
	Among sites within outbreak areas	1.421	0.254	0.002
	Among individuals within sites	558.594	99.746	

FIGURE 4 PCA biplots. (a) *Apanteles fumiferanae*, (b) *Glypta fumiferanae*. For each species, each dot represents an individual, and dot and inertia-ellipses colours represent the different outbreak areas shown in Figure 2 [Colour figure can be viewed at wileyonlinelibrary.com]



3.3 | Isolation by distance

For both parasitoid species, PCA-based genetic distances between each pair of individuals did not increase with geographic distance (Figure 5), despite very large distances between the most distant sites of our study area (i.e., >1,000 km; Figure 2a). As a result, we did not find evidence for IBD for either *A. fumiferanae* (Mantel $r = -0.0182$; $p = .3256$; Figure 5a) or *G. fumiferanae* (Mantel $r = -0.0532$; $p = .9216$; Figure 5b). This finding was confirmed using

the MEMGENE approach in that no significant Moran's eigenvectors were found for either species.

3.4 | Genetic diversity

The average expected heterozygosity (H_e) was $0.293 (\pm 0.015 \text{ SE})$ for *A. fumiferanae* and $0.291 (\pm 0.005)$ for *G. fumiferanae*, and the average allelic richness (A_r) was $1.650 (\pm 0.030)$ for *A. fumiferanae*

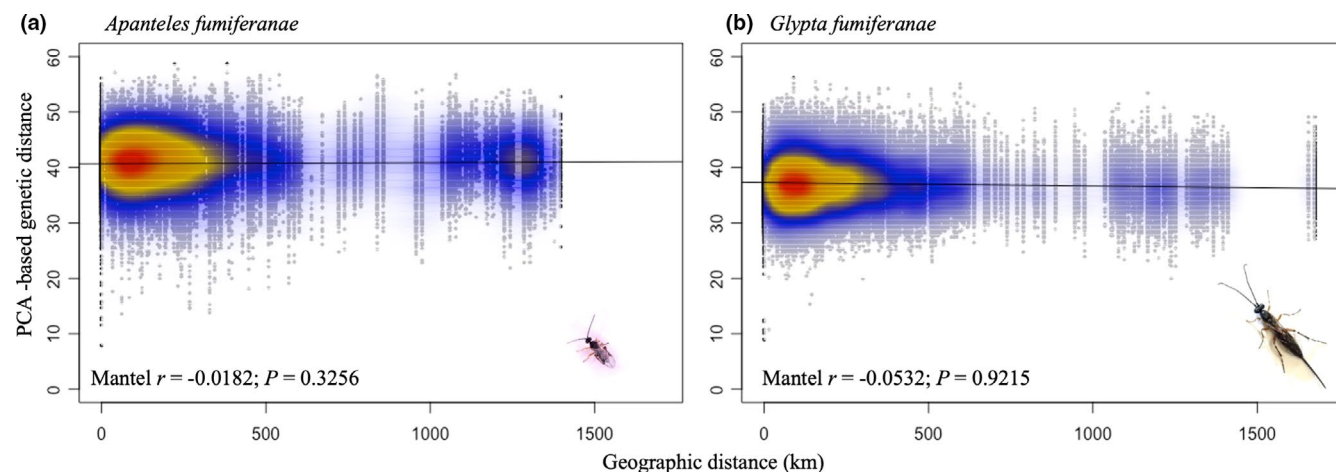


FIGURE 5 Isolation by distance (IBD) plots for each parasitoid species. Scatterplots show the relationships between interindividual genetic distance (PCA-based genetic distance) and geographic distances (in km) to test for the presence of IBD in (a) *Apanteles fumiferanae* and (b) *Glypta fumiferanae*. Colours represent the relative density of points, with warmer colours indicating higher densities. The black line indicates the fitted relationship between the two distance matrices using linear regression. The Mantel coefficient of correlation between geographic and genetic distances, as well as the associated p -value, is shown for each species. Using the MEMGENE approach, no significant Moran's eigenvectors were found for *A. fumiferanae* nor *G. fumiferanae* [Colour figure can be viewed at wileyonlinelibrary.com]

and $1.655 (\pm 0.008)$ for *G. fumiferanae*. Although differences in H_e and A_r were small between sites for both species (Table 1), we found significant negative correlations between H_e and latitude and between A_r and latitude for *A. fumiferanae* (Figure 6a, c). However, no significant correlations were found for *G. fumiferanae* (Figure 6b,d). Correlations using only sites with at least 10 genotyped individuals provided similar results (Figure 6).

4 | DISCUSSION

By attenuating food web fluctuations at large spatial scales through prey switching, higher-order predators can provide important ecosystem services that add stability to ecosystems experiencing natural perturbations (McCann & Rooney, 2009; Rooney et al., 2008). In insect food webs, parasitoids are often considered top predators, but despite their importance in the spatial trophic interactions that drive insect outbreaks (Royama et al., 2017), their dispersal abilities, a key component of prey switching (as described in terms of the bird-feeder effect; Eveleigh et al., 2007), are poorly understood (Cronin & Reeve, 2005), particularly in forest landscapes. In this study, we investigated the spatial genetic structure of two important spruce budworm parasitoids (*A. fumiferanae* and *G. fumiferanae*) to infer their population connectivity and identify potential barriers to dispersal. We undertook this investigation at a broad, provincial scale in different outbreak areas that have been experiencing a spruce budworm outbreak since ~2006. To our knowledge, this is the first large-scale spatial population genetic analysis of parasitoids in a continuous forest environment (Figure 2c). Our results allow us to infer the metapopulation structure of two key spruce budworm parasitoids, which is an important step toward understanding host-parasitoid spatiotemporal dynamics in the context of global change (Marrec et al., 2018).

4.1 | Genetic evidence of large-scale population connectivity

We did not find evidence for significant spatial genetic structure in either parasitoid species. This finding was confirmed using multiple methods. Using DAPC, a BIC minimum was always observed for $K = 1$ genetic cluster (Figure 3). Using AMOVA (Table 2) and PCA (Figure 4), we did not find any evidence for spatial genetic structure associated with the different outbreak areas or sampling locations, despite large distances between them (e.g., >1,000 km between outbreak areas I and VI; Figure 2a). Finally, we did not find any evidence for IBD using PCA-based genetic distances and geographic distances (Figure 5) with both Mantel tests and distance-based Moran's eigenvector maps. The absence of genetic structure in *A. fumiferanae* and *G. fumiferanae* suggests that parasitoid populations are functionally connected through high dispersal during spruce budworm outbreaks. If parasitoid populations fluctuated only at the local scale, without dispersing very far, we would have probably detected some form of genetic differentiation among populations from different outbreak areas. For example, different spatial patterns of genetic structure that correspond to their respective dispersal abilities have been identified in two competing parasitoids attacking the Glanville fritillary butterfly in the Åland Islands, Finland (Kankare et al., 2005).

The weak genetic structure we found for both parasitoid species matches the weak genetic structure recently described for the spruce budworm during the rising phase of the outbreak (Larroque et al., 2019). These results suggest that *A. fumiferanae* and *G. fumiferanae*, like the spruce budworm, have strong dispersal capacities and that both parasitoids can easily move toward areas of high host densities during the course of outbreak expansions. Other studies have also found evidence for high dispersal in parasitoid species. Several ichneumonid wasps have been introduced into Canadian forests to

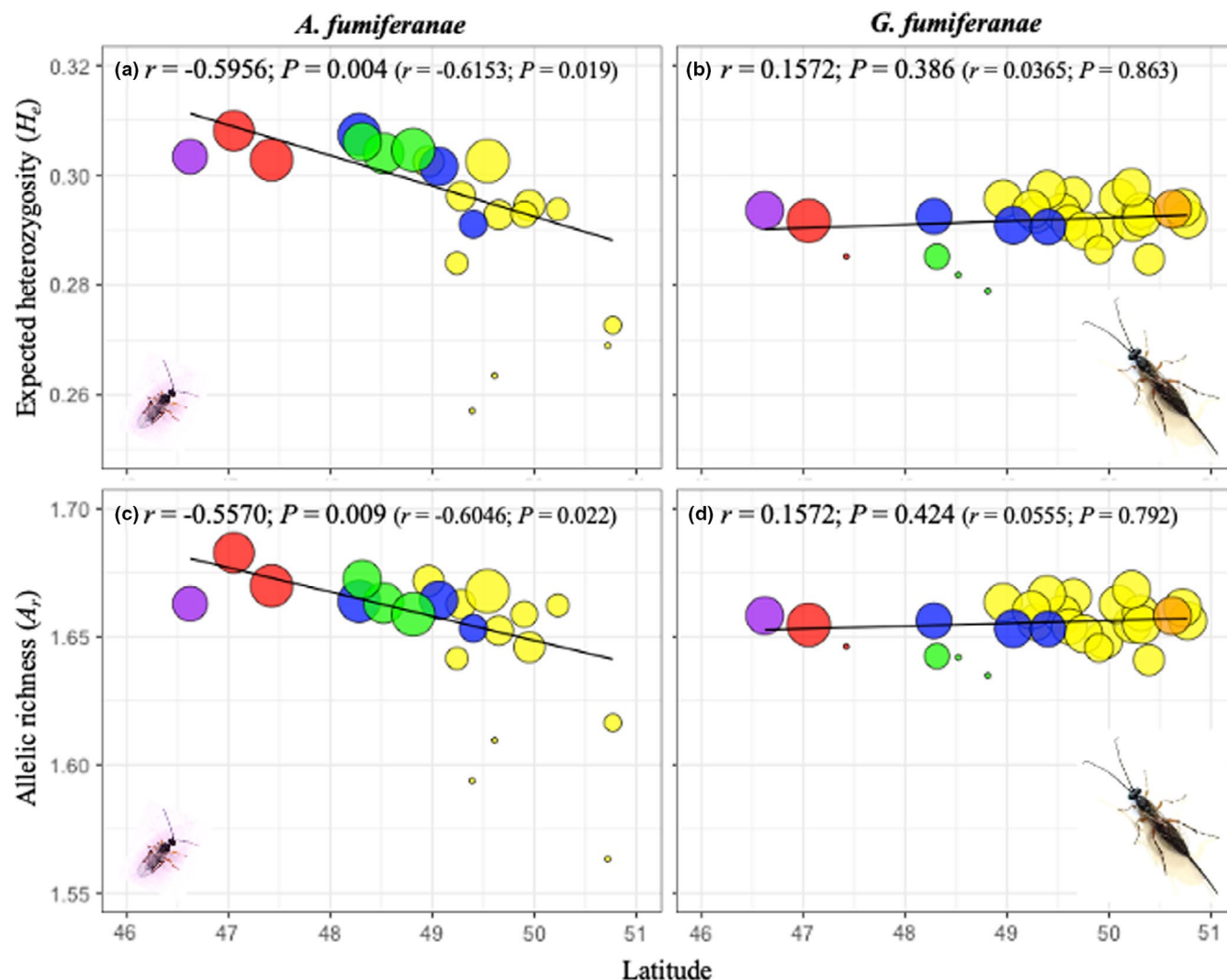


FIGURE 6 Testing the central-marginal hypothesis. Pearson correlations between mean expected heterozygosity (H_e) and latitude (a, b), and mean allelic richness (A_r) and latitude (c, d) for *A. fumiferanae* (a, c), and *G. fumiferanae* (b, d). Dot colours correspond to the different outbreak areas (see Figure 2a), and dot sizes are proportional to the number of parasitoid females (N) that were genotyped in order to estimate H_e and A_r values (Table 1). Pearson correlations and associated p -values are shown in parentheses for correlations limited to sites with at least 10 genotyped individuals [Colour figure can be viewed at wileyonlinelibrary.com]

control exotic defoliating pests and were able to disperse rapidly in contexts similar to that of our study. For example, *Olesicampe geniculatae* Quednau and Lim, a parasitoid of the mountain ash sawfly (*Pristiphora geniculata* [Hartig] [Hymenoptera: Tenthredinidae]) and *Lophyoplectus luteator* (Thunb.), attacking the European pine sawfly (*Neodiprion sertifer* [Geoffr.] [Hymenoptera: Diprionidae]) can disperse 50–140 km per year from their release point (Quednau, 1990; Rose, 1976; West et al., 1994), which is comparable to observed dispersal distances of the spruce budworm (Boulanger et al., 2017; Greenbank et al., 1980) and the distances between subsets of the sampled outbreak areas in our study (Figure 2a). Thus, in continuous forested landscapes, insect parasitoids may contribute in many cases to spatial synchrony in outbreak patterns by means of the bird-feeder effect (Eveleigh et al., 2007; McCann & Rooney, 2009).

The absence of genetic structure in *A. fumiferanae* and *G. fumiferanae* may also be the result of the highly connected and continuous

nature of northern coniferous forests that are composed mainly of spruce budworm host trees (Figure 2c). The population densities of both *A. fumiferanae* and *G. fumiferanae* increase with the local density of spruce budworm larvae (Miller, 1959, 1960), resulting in very high parasitoid densities during outbreaks that could reach many thousands of individuals per km^2 . With few barriers to dispersal and sufficiently high population densities, populations of these two parasitoid species could potentially remain well connected and effectively panmictic. This homogenization of genetic diversity could occur at the beginning of each large-scale outbreak. This dynamic has been hypothesized (James et al., 2015) and demonstrated (Larroque et al., 2019) for the spruce budworm itself within our study area. But contrary to what is observed for the spruce budworm, it is possible that parasitoid populations remain relatively high during endemic periods by switching to alternative hosts (Eveleigh et al., 2007; Krombein et al., 1979), especially for *A. fumiferanae* that

has been previously shown to be a generalist species (Smith et al., 2011). If so, we would hypothesize that parasitoid genetic differentiation will remain weak, irrespective of spruce budworm outbreak phases. This remains a question to be directly investigated.

4.2 | Support for the central-marginal hypothesis

The central-marginal hypothesis suggests that population size and effective dispersal should decrease toward species range margins (Eckert et al., 2008). Climate change is expected to produce more severe spruce budworm outbreaks in northern regions due to increased phenological synchrony with northern hosts (i.e., black spruce) and altered rates of parasitism due to reduced tree diversity (Bouchard & Auger, 2014; Legault & James, 2018; Pureswaran et al., 2018). Therefore, it is important to understand how different species of budworm parasitoids interact with this range margin.

We predicted that individuals from northern sites would exhibit greater genetic differentiation and lower genetic diversity than those from southern sites. Although we were unable to find evidence for significant genetic structure in *A. fumiferanae* and *G. fumiferanae* populations, we did find support for the central-marginal hypothesis in a significant negative relationship between genetic diversity (A_r and H_e) and latitude for *A. fumiferanae* (Figure 6a,c). We found no such support for *G. fumiferanae* (Figure 6b,d). This finding suggests that *A. fumiferanae* may be approaching its northern range limit in the northern sites sampled, whereas populations of *G. fumiferanae* may still be some distance from their northern limit. The comparatively lower genetic diversity in *A. fumiferanae* at the northern margin of the current outbreak (area V; Figure 2a) may be the result of reduced dispersal. Although other factors, such as lower alternative host densities and/or higher hyperparasitism, could also limit the abundance and connectivity of *A. fumiferanae* at higher latitudes (Legault & James, 2018). Future work should address these alternate explanations more thoroughly.

Predicting population dynamics in multitrophic systems in the context of climate change is challenging (Van Der Putten et al., 2004), but is necessary for assessing the resilience of forest ecosystems (Pureswaran, Neau, et al., 2018). Currently, little is known about the effects of range expansion and climate change on phenological synchrony among host trees, the spruce budworm, their parasitoids, and how these changes might affect outbreaks (but see Pureswaran, Neau, et al., 2018). Our findings suggest that the community of budworm-associated parasitoids may not respond uniformly in the short term to a climate-induced northward shift in the suitable range of the spruce budworm (Pureswaran et al., 2015). If key species of the spruce budworm parasitoid community, such as *A. fumiferanae*, are unable to track the northward movement of their host's range (Pureswaran, Neau, et al., 2018), regional changes in the overall strength of the birdfeeder effect due to alterations in trophic interactions within the spruce budworm-parasitoid-hyperparasitoid food web could have important consequences for forest resilience to future spruce budworm outbreaks.

4.3 | Methodological considerations and future directions

Using population genetic approaches to study cyclic and eruptive systems is challenging due to highly variable population dynamics and trophic relationships that depend on ecological context (Eveleigh et al., 2007; James et al., 2015). For instance, estimates of significant spatial genetic structure in the two parasitoid species we examined may vary with timing of sampling during an outbreak cycle (James et al., 2015; Larroque et al., 2019, 2020). Massive changes in population density of both the budworm and its parasitoids during outbreaks probably result in increased gene flow during epidemic periods (i.e., the timing of this study). One also expects reduced population genetic connectivity due to decreased population densities during endemic periods. Therefore, although we did find strong evidence of a high degree of population connectivity, this pattern may be transient and related to outbreak phase (James et al., 2015) and does not necessarily represent a fundamental aspect of parasitoid ecology.

Further, our estimates of parasitoid population genetic structure were based on parasitoids reared exclusively from spruce budworm larvae. We did not include parasitoids reared from alternative hosts, which both *A. fumiferanae* and *G. fumiferanae* may exploit during endemic phases of the spruce budworm cycle (Eveleigh et al., 2007; Krombein et al., 1979). The use of alternative hosts by these parasitoids may help maintain high parasitoid population levels during endemic phases and may help explain the high degree of population genetic connectivity we identified. Unfortunately, little is known about the role of nonbudworm hosts in maintaining endemic populations of parasitoids in boreal forest systems. More work in this area, specifically during endemic periods, is required to resolve these remaining uncertainties.

Finally, this study focused only one component (i.e., early larvae) of the very diverse spruce budworm-parasitoid food web (Eveleigh et al., 2007; Smith et al., 2011). Although the two species we studied are the most abundant within this food-web (Marrec et al., 2018), characterizing the genetic structure of a greater number of key parasitoid species would help understanding the complex host-parasitoid spatiotemporal dynamics in continuous forest environments. Specifically, it would be interesting to describe the genetic structure of other common primary parasitoid species, such as *Dirophanes maculicornis* (Stephens) (Hymenoptera: Ichneumonidae), *Meteorus trachynotus* (Vier.) (Braconidae) and *Smiditia fumiferanae* (Tothill) (Diptera: Tachinidae) (Marrec et al., 2018), and common hyperparasitoids such as *Stictopisthus* sp. Thomson (Ichneumonidae) and *Mesopolobus verditer* (Norton) (Hymenoptera: Chalcidoidea) (Eveleigh et al., 2007; Legault & James, 2018).

4.4 | Conclusions

Spatiotemporal patterns in the population dynamics of eruptive species have long fascinated ecologists (Elton, 1924; Pureswaran et al.,

2016; Schwartz et al., 2002). The role of natural enemies in driving these outbreaks is well acknowledged, although the population genetic consequences of outbreaks at higher trophic levels have seldom been examined. The consequences of outbreaks for population genetic structure at multiple trophic levels can improve our understanding of what drives spatial synchrony in these complex systems (James et al., 2011; Larroque et al., 2019; Schwartz et al., 2002). In this study, we characterized for the first time the spatial genetic structure of two widespread and important parasitoid species associated with the spruce budworm. Overall, we found evidence for very high genetic connectivity at large spatial scales (>350,000 km²) for both species. Prey switching by insect parasitoids, as described in terms of the birdfeeder effect (Eveleigh et al., 2007), probably accounts for this high degree of population connectivity and represents a plausible, although incomplete, explanation for the regional spatial synchronization of spruce budworm outbreaks.

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AUTHOR CONTRIBUTIONS

Simon Legault, Patrick M. A. James, Michel Cusson, and Jacques Brodeur conceptualized and designed the research. SL conducted the field sampling. Simon Legault and Julian Wittische analysed the data. Simon Legault wrote the manuscript. Patrick M. A. James, Julian Wittische, Michel Cusson, and Jacques Brodeur reviewed the manuscript.

DATA AVAILABILITY STATEMENT

VCF files of loci in approximate linkage equilibrium for *Apanteles fumiferanae* and *Glypta fumiferanae*, with their associated metadata, are available on Dryad (<https://doi.org/10.5061/dryad.rn8pk0pbj>).

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SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section.

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